

Rank Accounting for the Fixed-Misspecification Profile Hessian

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Purpose

The companion note `higher_order_discrepancy_misspec.tex` fixes the *mean* of the fitted discrepancy under fixed model and distributional misspecification: $2N\{v(\hat{x}) - v(x_0)\} \xrightarrow{d} Z'Q\Gamma Z$ with $\mathbb{E}[\hat{F}] = F_0 + n^{-1}[\text{tr}(Q\Gamma) + \beta_W] + o(n^{-1})$, the curvature-corrected generalized df $\text{tr}(Q\Gamma)$. This note does the complementary object: the *inertia* of the same $Q\Gamma$. The limiting law is a weighted mixture

$$2N\{v(\hat{x}) - v(x_0)\} \xrightarrow{d} \sum_{j=1}^m \lambda_j \chi_{1,j}^2, \quad \lambda_j = \text{eig}_j(Q\Gamma), \quad (1)$$

and the questions are: how many λ_j are nonzero (how many χ_1^2 terms are actually present), how many are positive, and exactly when the count collapses to the classical $df = m - p$. This is the analytic backing for the `spectrum_size` field of magmaan's `WeightedProfileRMSEAResult` / `WeightedProfileLRTResult`, which deliberately reports the size of the positive spectrum separately from the nominal model df because, as shown below, the two need not agree once an observed-bread or estimated-weight profile method is used. The single result is an inertia identity (Theorem 1); three corollaries read off when classical df survives (1), which channel breaks it (2), and what the broken case does to the magmaan summaries (4).

1 The profile Hessian

Fix the notation of the companion note. The moment vector is $s \in \mathbb{R}^m$ (the saturated statistics, $m = p^*(p^* + 1)/2$ for p^* observed variables), the

model is $\sigma(\theta)$, $\theta \in \mathbb{R}^p$ free after identification, with Jacobian $\dot{\sigma} = \partial\sigma/\partial\theta$ ($m \times p$, full column rank p) and curvature tensor $\ddot{\sigma}$. The fit minimizes a moment-quadratic discrepancy with an outer (data) metric V_o and a projection (model) metric W_p ; for a fixed-weight estimator the two coincide, $V_o = W_p = V$, while for a data-weighted estimator they differ. At the pseudo-true θ_0 write the population residual $e = \sigma(\theta_0) - s_\infty$, the Gauss–Newton part $A = \dot{\sigma}' W_p \dot{\sigma}$, the residual–curvature coupling $K = (W_p e)' \ddot{\sigma}$, and the discrepancy Hessian $B = A - K$. The profile Hessian is the curvature-corrected residual-weight matrix

$$Q = V_o - W_p \dot{\sigma} B^{-1} \dot{\sigma}' W_p \quad (= \tilde{U} \text{ of the companion note}), \quad (2)$$

in single-metric ($V_o = W_p = V$, ULS / fixed-weight GLS) or two-metric form ($V_o = V_0$ the data metric, $W_p = W_*$ the model metric, ML / GLS).

Remark (Why Q , not the naive sandwich). Q is exactly $\frac{1}{2}$ the Hessian of the profiled discrepancy $s \mapsto \min_\theta F(s, \theta)$ in the moment argument, by the envelope theorem $(\partial_s [\text{profiled } F]) = 2W_p r(\hat{\theta})$, then differentiate again using $\partial_s \hat{\theta} = B^{-1} \dot{\sigma}' W_p$. It is *not* the naive plug-in sandwich $(I - \dot{\sigma} B^{-1} \dot{\sigma}' W_p)' V_o (I - \dot{\sigma} B^{-1} \dot{\sigma}' W_p)$ obtained by linearizing $r(\hat{\theta})$ and squaring: that drops the $O_p(n^{-1})$ cross term between the leading and second-order parts of $r(\hat{\theta})$ and so disagrees with Q off H_0 (numerically $\|\text{sandwich} - Q\| = 0.95$ in the check below). The magmaan primitive stores Q , the envelope object, which is the correct one.

2 The inertia identity

Let $\text{In}(\cdot) = (n_+, n_0, n_-)$ denote the inertia (numbers of positive, zero, and negative eigenvalues) of a symmetric matrix. Define the *metric-gap form*

$$\tilde{A} = \dot{\sigma}' W_p V_o^{-1} W_p \dot{\sigma} \quad (p \times p), \quad (3)$$

which collapses to A in the single-metric case $V_o = W_p$.

Lemma 1 (Spectral split). *With $\tilde{G} = V_o^{-1/2} W_p \dot{\sigma}$ ($m \times p$, rank p),*

$$Q = V_o^{1/2} (I - \tilde{G} B^{-1} \tilde{G}') V_o^{1/2},$$

the column space $\text{col}(\tilde{G})$ is invariant under $I - \tilde{G} B^{-1} \tilde{G}'$, and

$$\text{spec}(I - \tilde{G} B^{-1} \tilde{G}') = \underbrace{\{1\}}_{\times(m-p)} \cup \{1 - \nu_j : j = 1, \dots, p\}, \quad \nu_j = \text{eig}_j(B^{-1} \tilde{A}). \quad (4)$$

Proof. On $\text{col}(\tilde{G})^\perp$ we have $\tilde{G}'u = 0$, so $(I - \tilde{G}B^{-1}\tilde{G}')u = u$: an eigenvalue 1 with multiplicity $m-p$. For $v = \tilde{G}w$, $(I - \tilde{G}B^{-1}\tilde{G}')\tilde{G}w = \tilde{G}(I - B^{-1}\tilde{G}'\tilde{G})w = \tilde{G}(I - B^{-1}\tilde{A})w$, since $\tilde{G}'\tilde{G} = \dot{\sigma}'W_pV_o^{-1}W_p\dot{\sigma} = \tilde{A}$. Thus $\text{col}(\tilde{G})$ is invariant and the operator acts there as $I - B^{-1}\tilde{A}$, with eigenvalues $1 - \nu_j$. The two invariant subspaces span $\mathbb{R}^m$. ν_j are real because $B^{-1}\tilde{A}$ is similar to the symmetric $\tilde{A}^{1/2}B^{-1}\tilde{A}^{1/2}$ ($\tilde{A} \succ 0$). $\square$

Theorem 1 (Rank accounting). *Q , $Q\Gamma$, and $I - \tilde{G}B^{-1}\tilde{G}'$ share the same inertia (Sylvester's law, congruence by $V_o^{1/2}$ and then by $\Gamma^{1/2}$, both nonsingular). Hence the limiting mixture (1) has*

$$\#\{\text{nonzero } \lambda_j\} = \text{rank}(Q\Gamma) = (m-p) + \#\{j : \nu_j \neq 1\}, \quad (5)$$

$$\#\{\text{positive } \lambda_j\} = n_+(Q\Gamma) = (m-p) + \#\{j : \nu_j < 1\}, \quad (6)$$

$$\#\{\text{negative } \lambda_j\} = n_-(Q\Gamma) = \#\{j : \nu_j > 1\}. \quad (7)$$

The classical chi-square picture is the special case in which all p of the ν_j equal 1, so the p model directions are exactly annihilated and only the $m-p$ residual directions survive.

Corollary 1 (When classical df survives). *$n_+(Q\Gamma) = \text{rank}(Q\Gamma) = m-p$ and $n_-(Q\Gamma) = 0$ if and only if $\nu_j = 1$ for all j , i.e. $B = \tilde{A}$. Sufficient is $B = A$ and $\tilde{A} = A$, that is $K = 0$ and $W_p = V_o$ on $\text{col}(\dot{\sigma})$. Both hold under correct specification ($e = 0$), and both hold for a model linear in θ with a fixed weight ($\ddot{\sigma} = 0 \Rightarrow K = 0$, $V_o = W_p \Rightarrow \tilde{A} = A$). When in addition $V_o = \Gamma^{-1}$ the surviving $m-p$ weights are all 1 and (1) is χ_{m-p}^2 .*

Corollary 2 (Two channels add directions). *ν_j departs from 1 through two independent, additive routes:*

- the **observed-bread (curvature) channel** $B \neq A$, i.e. $K \neq 0$: present whenever the residual meets a curved model, $\ddot{\sigma} \neq 0$. Factor and path models are nonlinear in θ , so $\ddot{\sigma} \neq 0$ generically and this channel is on for every wrong model. Absent for models linear in θ .
- the **estimated-weight (metric) channel** $\tilde{A} \neq A$, i.e. $W_p \neq V_o$: present whenever the projection metric differs from the data metric, which is exactly when the weight is tied to the model (ML: W_* at the fitted covariance versus V_0 at the population), or to the data and re-estimated. Absent for a fixed weight (ULS), where $V_o = W_p$ and only the curvature channel can act.

Each is $O(\|e\|)$ ($K \propto e$; $W_* - V_0 \propto e$) and vanishes at H_0 . ULS exposes one channel, ML/GLS both. This is the inertia-level sibling of the mean-level split in the companion note, where the curvature channel deforms $\text{tr}(U\Gamma) \rightarrow \text{tr}(\tilde{U}\Gamma)$ ($A \rightarrow B$) and the weight channel is β_W .

Corollary 3 (Sign of the extra weights; indefiniteness). *In the single-metric case write $\kappa_j = \text{eig}_j(A^{-1/2}KA^{-1/2})$, the curvature eigenvalues in the A-metric; then $\nu_j = (1 - \kappa_j)^{-1}$, and in the regular regime $B \succ 0$ ($\kappa_j < 1$),*

$$1 - \nu_j > 0 \iff \kappa_j < 0, \quad 1 - \nu_j < 0 \iff \kappa_j > 0.$$

So a negative residual-curvature direction adds a positive χ_1^2 weight, while a positive one adds a negative weight. Off H_0 with positive residual-curvature directions present, $Q\Gamma$ is therefore indefinite: the limiting law (1) is a difference of weighted χ_1^2 variables, not the noncentral χ^2 of the local-drift regime, and not even a nonnegative mixture.

3 Consequences for the magmaan summaries

Corollary 4 (Reading the fields). *`spectrum_size` = $n_+(Q\Gamma) = (m - p) + \#\{\nu_j < 1\} \geq m - p$, with equality exactly in the classical case of Corollary 1. The reported `bias_trace` sums only the positive λ_j , whereas the principled curvature-corrected df mean of the companion note is the signed trace $\text{tr}(Q\Gamma) = \sum_j \lambda_j$. When $n_-(Q\Gamma) > 0$ the two differ by $\sum_{\lambda_j < 0} |\lambda_j|$, so the positive-only `bias_trace` overcounts the noise relative to $\text{tr}(Q\Gamma)$. In the ULS check below the gap is 3.72 versus 2.26.*

This is a convention choice surfaced by the rank accounting, not a bug: truncating to the positive spectrum yields a conservative (larger) trace subtraction and a smaller RMSEA. The honest mean is the signed trace; the honest *count* of mixture terms is the positive spectrum. The two summaries answer different questions and the note recommends carrying both, which the struct already does.

Remark (Rank is brittle; effective rank tracks $\|e\|$). Equations (5)–(7) are exact-arithmetic statements. For any $e \neq 0$ the generic rank is $(m - p) + \text{rank}(\text{gap})$, but the extra eigenvalues emerge *continuously* as $O(\|e\|)$ from zero and the integer rank jumps only at $e = 0$. Near a correct model the extra weights are numerically negligible and the classical $m - p$ description is effectively right; far from it they are real and (1) is genuinely higher-rank and possibly indefinite. The reported `spectrum_size` should be read with a floor, as an effective rank that tracks the misfit, not as a discontinuous integer.

4 Numerical check

`profile_hessian_rank_check.py` (numpy only) fits a one-factor model to a two-factor population ($p^* = 6$, $m = 21$, classical $df = m - 2p = 9$; $\|e\| \approx 0.48$), reusing the construction of `higher_order_discrepancy_check.py`. It confirms, to machine precision: the spectral split (4) ($\max |\Delta| \sim 10^{-14}$); the factorization of Lemma 1; the Sylvester inertia equality $\text{In}(Q\Gamma) = \text{In}(Q) = \text{In}(I - \tilde{G}B^{-1}\tilde{G}')$; and the counts (5)–(7) against a direct eigendecomposition of $Q\Gamma$.

estimator	channels	df	$\text{In}(Q\Gamma) = (n_+, n_0, n_-)$	n_+	$\text{tr}(Q\Gamma)$	$\sum_{\lambda > 0} \lambda$
ULS (analytic K)	curvature	9	(13, 7, 1)	13	2.26	3.72
ML (two metric)	curvature+weight	9	(17, 1, 3)	17	1.24	10.61

Both sit strictly above classical df , ULS by the one curvature channel (four extra positive, one negative weight) and ML by both channels (eight extra positive, three negative). A residual-shrink scan pulling the population onto the fitted model ($\alpha : 1 \rightarrow 0$) drives the effective rank of both down to 9 and the negative count to 0 as $\|e\| \rightarrow 0$, with the largest extra weight shrinking like $O(\|e\|)$, confirming Corollaries 1 and 2 and the brittleness remark. The naive-sandwich check of Remark 1 reproduces $\|\text{sandwich} - Q\| = 0.95 \neq 0$.

What remains

The identity is structural and estimator-generic (single- and two-metric), so it already covers ULS/GLS/WLS/DWLS and covariance ML at the level of Q . What it does not yet do, and what a downstream consumer would need, is (i) an *analytic* count of $\#\{\nu_j < 1\}$ from model structure rather than an eigendecomposition, which would require signing K and the metric gap a priori; (ii) the mean-structure and FIML/ML2S two-metric Q , where $\tilde{A}$ and B acquire the missing-data and mean blocks; and (iii) the categorical (u, γ) first-stage expansion, where the first-stage influence adds its own directions to Γ on top of the two channels here. Those are the open profile-Hessian items in `docs/backlog/todo.md`; this note settles the inertia bookkeeping they all build on.